Hochschule Luzern

Master of Science in Applied Information and Data Science (MScIDS)

Preliminary Study

Heart Disease

Submitted as part of the module

W.HEA01 – Data Science in Healthcare

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1. INTRODUCTION & MOTIVATION

More than one third of global deaths are attributable to cardiovascular diseases (CVDs). In 2022, this corresponded to 19.8 million deaths worldwide. Of these, 85% were caused by heart attacks and strokes (*Cardiovascular Diseases (CVDs)*, 2025). According to the Bundesamt für Statistik (2025), cardiovascular diseases account for between 21% and 37% of all deaths in Switzerland, depending on age. Overall, they generate more than CHF 20 billion annually in direct and indirect healthcare costs (Rosemann et al., 2024). Cardiovascular diseases therefore represent a major societal and health-economic burden.

The early identification of patterns and risk factors, before a serious cardiac condition develops, could enable many individuals to live longer and healthier lives while simultaneously reducing pressure on the healthcare system.

This study therefore investigates, based on a heart disease dataset, the extent to which clinical and demographic variables can be used to predict heart disease. Relationships between variables will be analyzed and appropriate modeling approaches for risk prediction will be evaluated. The aim is to identify relevant risk factors and, based on these findings, to derive an interpretable risk score.

2. DATA DESCRIPTION

For the present analysis, the "Heart Disease" dataset from the UCI Machine Learning Repository is used (Janosi et al., 1989). The originally published dataset comprises 76 attributes and is based on four regional sub-datasets: Cleveland, Hungary, Switzerland and VA Long Beach. In this study, the reduced version with 14 variables commonly used in the literature is analyzed. These 14 variables are contained in all four regional sub-datasets. The four sub-datasets comprise a total of 920 observations (Cleveland: 303, Hungary: 294, Switzerland: 123, VA Long Beach: 200). The target variable ("num") indicates the presence of heart disease. The predictor variables include both continuous and categorical features. These comprise:

- Age (age)
- Sex (sex)
- Chest pain type (cp)
- Resting blood pressure (trestbps)
- Serum cholesterol in mg/dl (chol)
- Fasting blood sugar > 120 mg/dl (fbs)
- Resting electrocardiographic results (restecg)
- Maximum heart rate achieved (thalach)
- Exercise-induced angina (exang)
- ST depression induced by exercise relative to rest (oldpeak)
- Slope of the peak exercise ST segment (slope)
- Number of major vessels (0–3) colored by fluoroscopy (ca)
- Thalassemia (thal)

The dataset is provided in structured tabular form. Each row represents one individual and each column represents one variable. A detailed description of all variables is provided in the appendix.

3. RESEARCH QUESTIONS

The project aims to resolve following questions:

Main research question:

- How can an interpretable probability-based risk score for heart disease be developed based on clinical and demographic characteristics?

Sub questions:

- Which predictors show the strongest relationship with the presence of heart disease?
- Which subset of predictors should be included in the risk score?
- How should these predictors be weighted?

4. METHODOLOGICAL PLAN

The methodological steps are organized as tasks and summarized in a project timeline in the appendix.

4.1. Data Preparation

The four regional sub-datasets (Cleveland, Hungary, Switzerland and VA Long Beach) will first be combined into a single dataset. Before merging, variable names and coding schemes will be checked to ensure consistency across regions. The target variable will be defined as binary, indicating the presence or absence of heart disease.

Missing values will be examined for each variable. Continuous predictors will be imputed using median values, while categorical predictors will be imputed using the most frequent category where appropriate. After imputation, categorical variables such as chest pain type, resting ECG results, thalassemia and ST slope will be transformed using one-hot encoding. Continuous variables will be standardized to allow for regularized modeling and comparability of coefficients.

Finally, the dataset will be divided into training and test sets using stratified sampling to maintain the distribution of the outcome variable.

4.2. Exploratory Analysis

Descriptive statistics will be calculated for all variables in order to understand their distributions and potential irregularities. For continuous predictors, measures such as mean, median and standard deviation will be reported and their distributions will be visually inspected. For categorical variables, frequency distributions will be examined.

Relationships between predictors and the target variable will be explored using correlation analysis, group comparison and visualization techniques. In addition, potential multicollinearity among continuous variables will be assessed using the Variance Inflation Factor (VIF). This step will help to identify redundant predictors before model estimation.

4.3. Analytical Strategy

Logistic regression will serve as the primary modeling approach due to its interpretability and suitability for binary outcomes. To reduce the risk of overfitting and to identify relevant predictors, L1-regularized logistic regression (LASSO) combined with k-fold cross-validation will be applied.

In addition, a gradient boosting classifier will be implemented as a performance benchmark. Hyperparameters will be tuned using cross-validation to ensure a fair comparison between models.

Model performance will primarily be evaluated using ROC-AUC, complemented by precision, recall and F1-score. Because the final objective is to produce meaningful probability estimates, model calibration will also be assessed using calibration curves and the Brier score.

The final risk score will be defined as the predicted probability obtained from the logistic regression model. The influence of individual predictors will be interpreted through model coefficients, supported by SHAP analysis where appropriate.

5. EXPECTED CONTRIBUTION

This study integrates machine learning and interpretable modeling to derive a probability-based risk score for heart disease. By combining predictive performance with transparent coefficient interpretation, the analysis provides a structured and reproducible framework for data-driven cardiovascular risk assessment.

6. POTENTIAL CHALLENGES

Differences in data structure and missing value patterns across regional sub-datasets may affect model stability. The moderate sample size limits model complexity and increases the importance of regularization and cross-validation. A trade-off between predictive performance and interpretability must be carefully managed.

LITERATURE

Bundesamt für Statistik. (2025, Dezember 22). *Spezifische Todesursachen*. https://www.bfs.admin.ch/content/bfs/de/home/statistiken/gesundheit/gesundheitszustand/sterblichkeit-todesursachen/spezifische.html

Cardiovascular diseases (CVDs). (2025, Juli 31). World Health Organization. https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)

Janosi, A., Steinbrunn, W., Pfisterer, M., & Detrano, R. (1989). *Heart Disease* [Dataset]. UCI Machine Learning Repository. https://doi.org/10.24432/C52P4X

Rosemann, T., Bachofner, A., & Strehle, O. (2024). Kardiovaskuläre Erkrankungen in der Schweiz – Prävalenz und Versorgung. *PRAXIS*, *113*, 57–66.

USE OF AI

Generative AI tools were used to support the project in two ways. First, language models were used to assist with wording and structural feedback during the preparation of this preliminary study. Second, coding-oriented large language models will be used during the main study to support the implementation of data preprocessing, exploratory analysis and machine learning models in Python.

All generated code will be critically reviewed, tested and adapted by the authors. The analytical design, methodological decisions and interpretation of results remain the responsibility of the authors.

APPENDIX

APPENDIX 1 - Detailed data

Var	Description	Type	Units / Value Meaning
age	Age of patient	Numeric – Continuous	Years
trestbps	Resting blood pressure	Numeric – Continuous	mm Hg
chol	Serum cholesterol	Numeric – Continuous	mg/dl
thalach	Maximum heart rate achieved	Numeric – Continuous	Beats per minute (bpm)
oldpeak	ST depression induced by exercise relative to rest	Numeric – Continuous	Relative ECG depression value
ca	Number of major vessels colored by fluoroscopy	Numeric – Discrete	Count (0–3 vessels)
sex	Sex of patient	Categorical – Binary (Nominal)	1 = male; 0 = female
fbs	Fasting blood sugar > 120 mg/dl	Categorical – Binary (Nominal)	1 = true (>120 mg/dl); 0 = false
exang	Exercise induced angina	Categorical – Binary (Nominal)	1 = yes; 0 = no
cp	Chest pain type	Categorical – Multiclass (Nominal)	1 = typical angina; 2 = atypical angina; 3 = non-anginal pain; 4 = asymptomatic
restecg	Resting electrocardiographic results	Categorical – Multiclass (Nominal)	0 = normal; 1 = ST-T wave abnormality; 2 = left ventricular hypertrophy
thal	Thalassemia test result	Categorical – Multiclass (Nominal)	3 = normal; 6 = fixed defect; 7 = reversible defect
slope	Slope of peak exercise ST segment	Categorical – Ordinal	1 = upsloping; 2 = flat; 3 = downsloping
num	Diagnosis – heart disease severity **(target)**	Numeric – Ordinal	0 = upsloping; 1 - 4 = increasing severity (binarised)
source	Originating cohort	Categorical – Multiclass (Nominal)	1 = Cleveland 2 = Hungary 3 = Long Beach VA 4 = Switzerland

APPENDIX 2 - Project timeline

		Feb	Mar			
		KW9	KW10	KW11	KW12	KW13
Preliminary Study						
Search for topic, busines case & datasources	all					
Submission Preliminary Study	all		♦			
Main Project						
Data Preparation	all					
EDA	all					
Model development & evaluation	all					
Interpretation of results & risk score derivation	all					
Final Presentation	all					♦

APPENDIX 3 – Team Agreement

The goal of this Team Agreement is to facilitate your group in establishing a solid basis as a project team that you can rely on during the entire project for coordinating your teamwork and for dealing with problems effectively, in the case that they occur.

1. Contact information

	Group name	**Topic**	
	Member	E-mail	Phone
1	Sandra Deck	sandra.deck@stud.hslu.ch	076 490 57 95
2	Joshua Hügli	joshua.huegli@stud.hslu.ch	079 266 14 81

2. Defining your individual and team goals in this project

Please state your team goal for this project:

Our goal as a team is to collaborate effectively and ensure that we deliver a high-quality project that meets all course requirements.

Deliverables:

Member	Deliverables
2	Data preparation, exploratory data analysis, and support in model implementation
1	Literature review, research question development, and methodological design
1, 2	Model development, result interpretation, visualization, and preparation of the final report and presentation

3. Defining rules of conduct

Please specify consequences for a member of your team who does not adhere to deadlines, fails to attend meetings, or does not respond to e-mails/calls:

If someone is unable to meet a deadline or attend a meeting, they should inform the other member as early as possible.

Please state what steps you will take when you are not satisfied with each other's contributions:

Step 1: Short discussion to clarify expectations and current tasks.

Step 2: If necessary, tasks will be adjusted or redistributed.

Step 3: If the issue continues, the instructor can be contacted for advice.

Once you have drafted your Team Agreement, meet and discuss any edits or changes that are warranted.